



FITHUB: AN ADVANCED DISTRIBUTED APP ECOSYSTEM FOR FITNESS COMPLEX MANAGEMENT

CE405.3

FINAL REPORT

Project Team Identification

Project Member 01:

- Name: N.W.P.P.P. Nanayakkara
- Index Number: 29366
- Degree Programme: BSc (Hons) in Computer Systems Engineering

Project Member 02:

- Name: W.I.J.M.R.K. Warnasuriya
- Index Number: 28009
- Degree Programme: BSc (Hons) in Computer Systems Engineering

Contents

CHAPTER 01: INTRODUCTION	4
1.1 Existing Systems and Problem Definition	4
1.2 Project Aims and Objectives.....	4
1.3 Scope of the Project	5
CHAPTER 02: REQUIREMENTS ELICITATION AND SPECIFICATION	5
2.1 Requirement Gathering Techniques.....	5
2.2 Functional Requirements.....	5
2.3 Non-Functional Requirements.....	6
2.4 Structural Application Features	6
CHAPTER 03: SYSTEM DESIGN AND USER INTERFACES	7
3.1 Use Case Diagram	7
3.2 High-Level Diagram.....	8
3.3 ER Diagram.....	8
3.4 Developed User Interface Analysis	9
Owner Dashboard View Interfaces	9
3.4.1 Welcome Screen and Role Selection Portal	9
3.4.2 Owner Login Credentials Panel Interface	10
3.4.3 Two-Step Verification OTP Panel Interface	10
3.4.4 Gym Registration and Location Mapping Interface	11
3.4.5 Password Recovery Reset Request Interface	11
3.4.6 Gym Owner Administrative Command Hub Interface	12
3.4.7 Owner Navigation Drawer Menu.....	12
3.4.8 Create Membership Plan Interface	13
3.4.9 Add New Staff Registration Form	13
3.4.10 Add New Gym Member Interface.....	14
3.4.11 Club Member Directory Dashboard	14
3.4.12 Gym Staff Registry Subsystem	15
3.4.13 Detailed Staff Profiles View.....	15
Trainer Dashboard View Interfaces	16
3.4.14 Personal Trainer Hub Home Dashboard.....	16
3.4.15 Personal Trainer Assigned Clients Directory	16
3.4.16 Goal-Oriented Workout Assignment Roster.....	17
3.4.17 Custom Exercise Routine Building Canvas.....	17
3.4.18 AI-Suggested Workout Matrix Validation Canvas	18
3.4.19 Goal-Oriented Dietary Assignment Roster.....	18
3.4.20 AI-Suggested Dietary Target Matrix Preview	19

3.4.21 Personal Trainer Master Booking Schedule.....	19
3.4.22 Detailed Club Member Biographical Profile	20
3.4.23 Club Member Subscription Plan Management View	20
3.4.24 Club Member Attendance History Tab	21
3.4.25 Member Assigned Workout Plans Feed.....	21
3.4.26 Member Dietary Target Plans Feed	22
Member Dashboard View Interfaces.....	22
3.4.27 Gym Member Landing Hub Workspace	22
3.4.28 Member Assigned Workouts Registry.....	23
3.4.29 Workout Routine Activity Breakdown.....	23
3.4.30 Interactive AI Squat Coach View	24
3.4.31 Dietary Program Nutritional Target Feed	24
3.4.32 Member Growth Analytics Dashboard	25
3.4.33 Gym Announcements Feed Panel	25
3.4.34 Automated WhatsApp Onboarding Credential Notification Subsystem	26
Database	26
CHAPTER 04: METHODOLOGY AND IMPLEMENTATION	28
4.1 Development Methodology.....	28
4.2 Technologies and Tools Used	28
4.3 Future Implementation	28
CHAPTER 05: PLATFORM GOVERNANCE AND DELIVERY.....	29
5.1 Individual Contribution Statement.....	29
5.2 GitHub Commit History.....	29
5.3 Project Source Code Deployment Vector	30
REFERENCE LIST	31

CHAPTER 01: INTRODUCTION

1.1 Existing Systems and Problem Definition

Traditional fitness centers struggle with notable operational inefficiencies due to decentralized workflows. Administrative tasks often rely on disjointed spreadsheet records. Manual bookkeeping methods, or isolated file systems. Honestly, these outdated systems just don't provide real-time data synchronization across connected devices.

Data segregation creates real challenges for fitness instructors, and they struggle to monitor how effectively members actually stick to their routines; plus, updating physiological data in real time becomes kind of a hassle. Fixed routines. Which members end up with, don't adapt based on their performance variations or how they execute things individually. Anyway, current security setups fail to incorporate modern validation techniques.

1.2 Project Aims and Objectives

This engineering project aims to design and deploy a secure, cross-platform distributed application ecosystem named **FitHub**. **FitHub** is being organized to fix missing work gaps since **FitHub** creates a digital workspace for fitness center operators, physical training professionals, and active members.

The primary technical objectives include:

- Developing a zero-trust multi-role access framework integrated with multi-factor cellular authentication gateways.
- Engineering reactive graphical dashboard modules that automatically render configurations based on verified user permissions.
- Deploying an edge-computed computer vision engine capable of executing real-time skeletal alignment tracking for exercise form correction.
- Constructing an enterprise-grade database pipeline to manage financial transaction processing and immediate notice distribution.

1.3 Scope of the Project

It is conceived and maintained end-to-end with full-stack software development, from database modeling, responsive front end framework, back end hosting in cloud infrastructure, as well as isolation of models for three primary user classes: Gym Owners, Personal Trainers, and Gym Members.

Delivery of the system with consistent cross platform performance, high-security validation and low latency for all distributed concurrent clients.

CHAPTER 02: REQUIREMENTS ELICITATION AND SPECIFICATION

2.1 Requirement Gathering Techniques

To define requirements, a multi-tier elicitation strategy was used, which included two initiatives. First, organized virtual interviews were held with regional facility managers to map the existing corporate accounting systems and establish structural operational constraints. Second, a series of observational walkthroughs with personal fitness coaches identified scheduling challenges and bottlenecks in manual processes.

Additionally, digital surveys were distributed to existing gym members, capturing user choices regarding real-time physiological data tracking visuals and interface interactivity.

2.2 Functional Requirements

- **FR-1.1 (Multi-Role Authentication Gateway):** Authentication by secure email paths and secondary cellular multi factor short code challenges must be supported.
- **FR-1.2 (Dynamic State-Driven Views):** All valid authorization headers will be identified automatically, and users will be directed to their appropriate role dashboard.
- **FR-1.3 (AI-Assisted Plan Generation):** The architecture should create exercise plans and nutrition plans that are tailored to the client's goals and metrics, using the AI tool.
- **FR-1.4 (Edge Based Form Tracking):** The platform should capture the 2D form of the skeleton in real time with the use of camera arrays on the local device and evaluate the alignment of the motion.

- **FR-1.5 (Administrative Data Hub):** The application must process commercial transactions, display financial analytics summaries, and broadcast global notice bulletins.

2.3 Non-Functional Requirements

- **NFR-2.1 (Data Transport Security):** The encryption of all database operations must be done at rest and at transit with the most modern secure connection protocols, with support of granular back-end verification constraints.
- **NFR-2.2 (Synchronization Responsiveness):** The execution time for state changes across terminal application nodes should be within 1.5 seconds with the help of persistent live snapshot listeners.
- **NFR-2.3 (Resource Efficiency Limits):** Power up and reach structural responsiveness within 2.5 seconds on consumer mobile processor chips, with low battery consumption when using the camera..

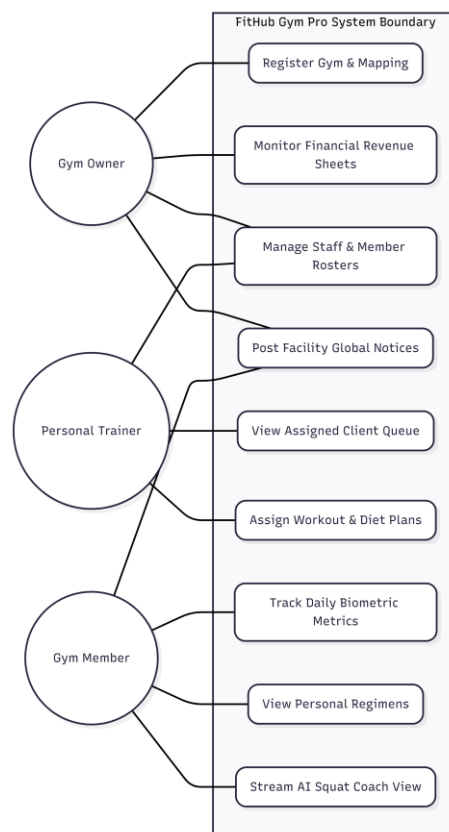
2.4 Structural Application Features

FitHub is also equipped with tracking systems, a multi-paragraph viewer of compliance policies, a developer validation access control system, and a custom debugging menu that can be accessed from its app bar shell. It supports smooth switching between Owner, Trainer, and Member execution modes for cross-role verification of every situation without interfering with database isolation.

CHAPTER 03: SYSTEM DESIGN AND USER INTERFACES

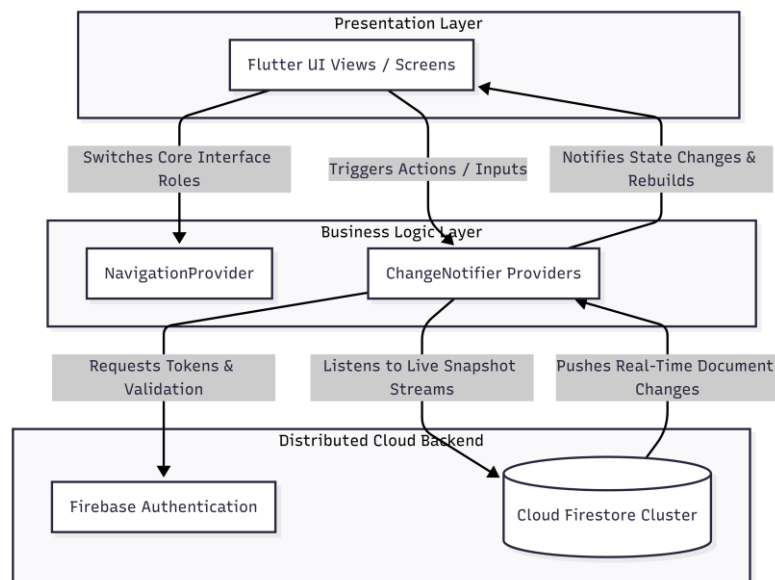
3.1 Use Case Diagram

Platform context is the context in which permissions are enforced, using different administrative roles. The Gym Owner is the main administrative body, and has all the privileges to keep track of the amount of money, adjust the staff schedule and post Gym announcements. The Personal Trainer manages client processes, arranges schedule blocks and provide workout and diet matrices. The Gym Member uses performance monitoring systems, logs daily exercise programs, reads notice feeds and uses interactive real-time movement form feedback systems.



3.2 High-Level Diagram

The system architecture is based on the concept of a reactive system controlled by change notifier. The presentation layer directly communicates with business logic handlers, which communicate with clusters of cloud storage and clouds of databases. This way state and interface processing are kept apart, preventing corrupted memory buffers due to network problems or invalid user actions.



3.3 ER Diagram

The system uses a fast, document-oriented data model optimized for distributed synchronization:

- **gyms_master collection:** Tracks facility location coordinates, unique reference tokens, and primary email profiles.
- **profiles_user collection:** Stores specific biological records, historical weight timelines, and verified role permission tags.
- **schedules_fitness collection:** Controls assigned exercise sequences, repetition counts, and specialized dietary targets.



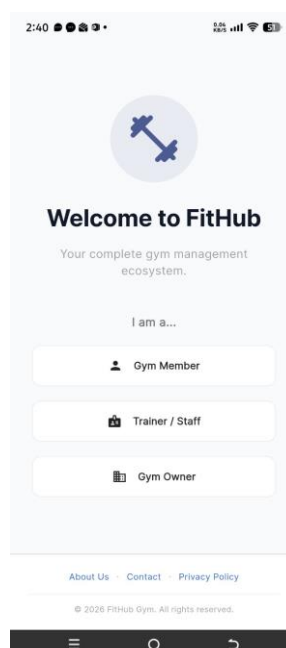
3.4 Developed User Interface Analysis

The system's user interface is constructed using a refined mobile dashboard palette designed to maximize visual clarity during intense training environments. The complete, ordered inventory of system views is mapped below:

Owner Dashboard View Interfaces

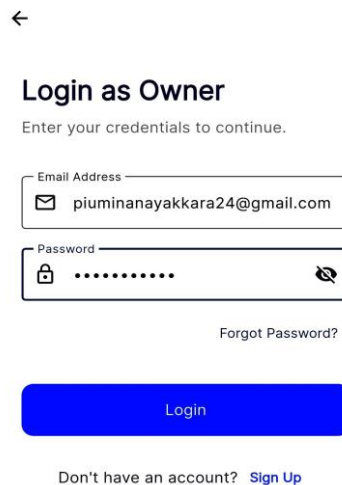
3.4.1 Welcome Screen and Role Selection Portal

This primary entry screen handles initial application routing. The layout features structural buttons that classify the session user context as a Gym Member, Trainer / Staff, or Gym Owner.



3.4.2 Owner Login Credentials Panel Interface

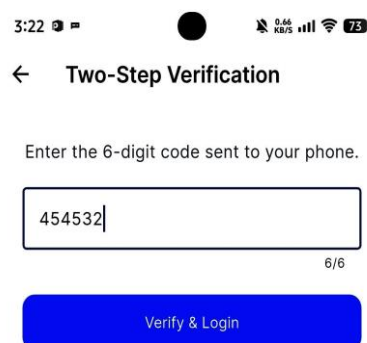
This secure panel captures administrative authentication profiles. It provides email validation fields and an interactive password visibility toggle connected to serverless token generation engines.



A mobile application login screen titled "Login as Owner". At the top left is a back arrow. Below the title is the instruction "Enter your credentials to continue." There are two input fields: "Email Address" containing "piuminanayakkara24@gmail.com" and "Password" with masked dots and a visibility toggle icon. Below the password field is a "Forgot Password?" link. A large blue "Login" button is centered below the fields. At the bottom, there is a link "Don't have an account? Sign Up".

3.4.3 Two-Step Verification OTP Panel Interface

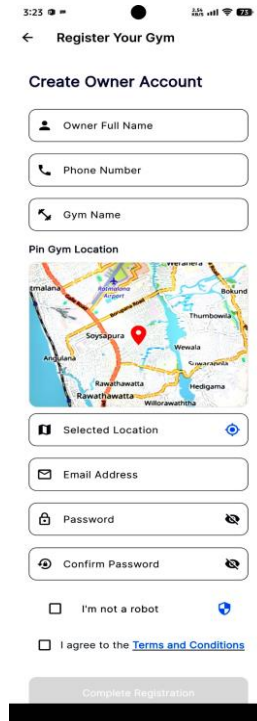
Following primary password verification, this mandatory verification layer challenges users to provide a 6-digit short-code sent to their verified mobile device.



A mobile application screen for "Two-Step Verification". The status bar at the top shows the time 3:22, a battery icon, and network indicators. A back arrow is at the top left. The title "Two-Step Verification" is at the top. Below it is the instruction "Enter the 6-digit code sent to your phone." There is a text input field containing "454532" with a "6/6" character count indicator to its right. A large blue "Verify & Login" button is centered at the bottom.

3.4.4 Gym Registration and Location Mapping Interface

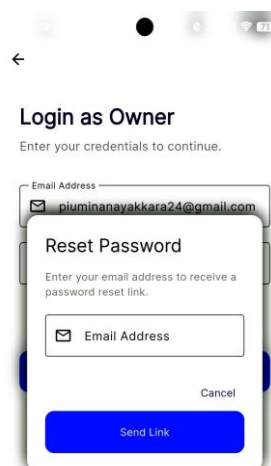
This view handles onboarding workflows for new gym configurations. It includes text validation boundaries alongside an interactive geospatial coordinate selector to confirm the physical gym location.



The screenshot shows a mobile app interface for registering a gym. At the top, there's a status bar with the time 3:23 and battery level 62%. Below the status bar is a back arrow and the title "Register Your Gym". The main section is titled "Create Owner Account". It contains three input fields: "Owner Full Name", "Phone Number", and "Gym Name". Below these is a section titled "Pin Gym Location" which includes a map showing a location in Sri Lanka. Under the map is a "Selected Location" button with a location pin icon. Below the map are four more input fields: "Email Address", "Password", "Confirm Password", and a checkbox for "I'm not a robot". At the bottom, there's a checkbox for "I agree to the Terms and Conditions" and a "Complete Registration" button.

3.4.5 Password Recovery Reset Request Interface

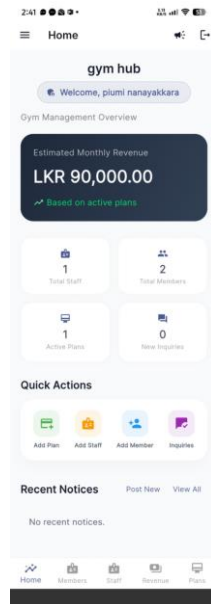
This overlay modal processes self-service account recovery workflows. It routes automated credential restoration payloads to verified administrative inboxes.



The screenshot shows a mobile app interface for logging in as an owner. At the top, there's a status bar with the time 3:23 and battery level 62%. Below the status bar is a back arrow and the title "Login as Owner". Under the title is the text "Enter your credentials to continue." Below this is an "Email Address" input field with the email address "piuminanayakkara24@gmail.com" entered. A modal overlay titled "Reset Password" is shown in the foreground. The modal contains the text "Enter your email address to receive a password reset link." and an "Email Address" input field. At the bottom of the modal are two buttons: "Cancel" and "Send Link".

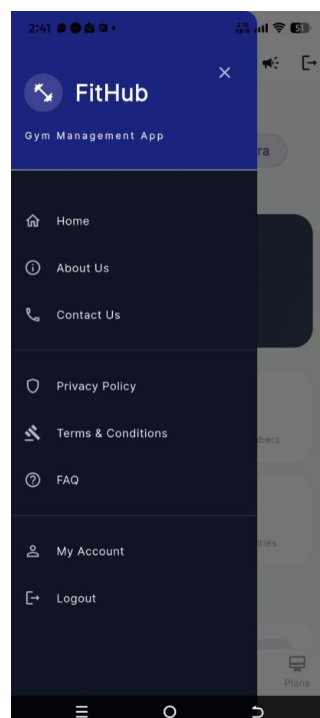
3.4.6 Gym Owner Administrative Command Hub Interface

The home workspace for verified gym administrators. It displays monthly business records (*LKR 90,000*) next to reactive metrics summaries and operational quick action vectors.



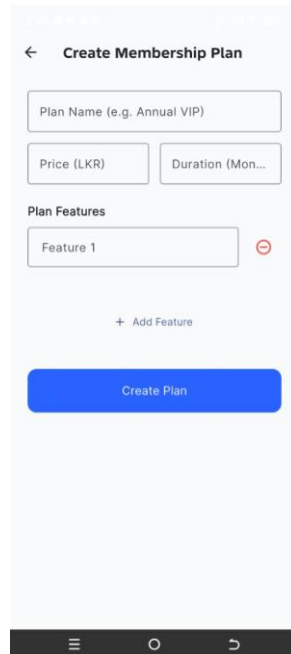
3.4.7 Owner Navigation Drawer Menu

A sidebar navigation interface allowing immediate routing between application subsystems, legal policies, and active session logouts.



3.4.8 Create Membership Plan Interface

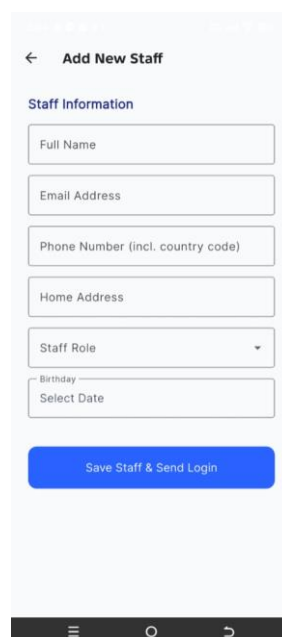
An administrative form used to define new package parameters, pricing details, monthly durations, and custom feature tags.



The screenshot shows a mobile application interface for creating a membership plan. At the top, there is a back arrow and the title "Create Membership Plan". Below the title, there is a form with the following fields: "Plan Name (e.g. Annual VIP)", "Price (LKR)", and "Duration (Mon...)". Under these fields, there is a section titled "Plan Features" with a text input field labeled "Feature 1" and a red minus icon to its right. Below the "Feature 1" field, there is a "+ Add Feature" button. At the bottom of the form, there is a large blue button labeled "Create Plan". The bottom of the screen shows a standard Android navigation bar with three icons: a hamburger menu, a circle, and a back arrow.

3.4.9 Add New Staff Registration Form

An input subsystem allowing gym owners to register employee information, assign specific system roles, and trigger credential distributions.



The screenshot shows a mobile application interface for adding new staff. At the top, there is a back arrow and the title "Add New Staff". Below the title, there is a section titled "Staff Information" with the following fields: "Full Name", "Email Address", "Phone Number (incl. country code)", "Home Address", "Staff Role" (a dropdown menu), and "Birthday" (a date picker labeled "Select Date"). At the bottom of the form, there is a large blue button labeled "Save Staff & Send Login". The bottom of the screen shows a standard Android navigation bar with three icons: a hamburger menu, a circle, and a back arrow.

3.4.10 Add New Gym Member Interface

A comprehensive setup screen capturing candidate biographical metrics, physical attributes, and membership registration baselines.

Add New Member

Personal Information

Full Name

Email Address

Phone Number (incl. country code e...

Home Address

Physical Details

Height (cm) Weight (kg)

Age Birthday Select Date

Membership Details

Membership Plan Select a Plan

Start Date Select Date

Save Member & Send Login

3.4.11 Club Member Directory Dashboard

A directory roster displaying active gym profiles along with individual registration IDs, quick contact actions, and plan expiration tracking.

Members

Search by name, ID or phone...

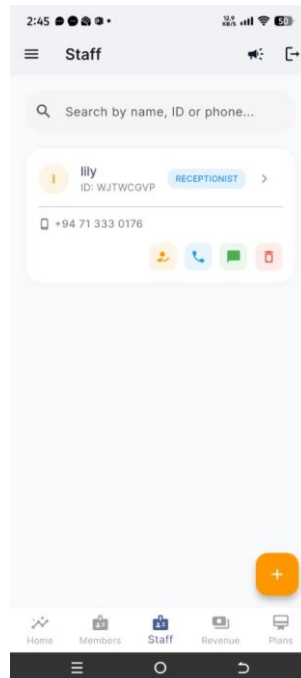
upali ID: BFNPKLN

0712603521 Plan Ends: 2027-06-18

Home Members Staff Revenue Plans

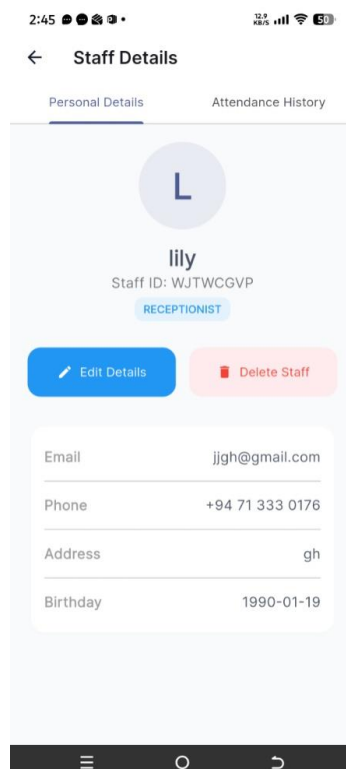
3.4.12 Gym Staff Registry Subsystem

A personnel dashboard summarizing current active staff records, corresponding roles, and administrative data management options.



3.4.13 Detailed Staff Profiles View

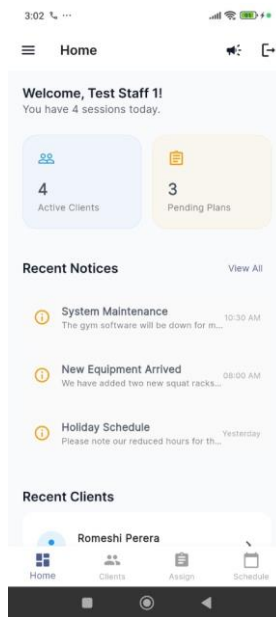
An individual profile page grouping detailed human resource records, biographical information, and structural staff attendance logs.



Trainer Dashboard View Interfaces

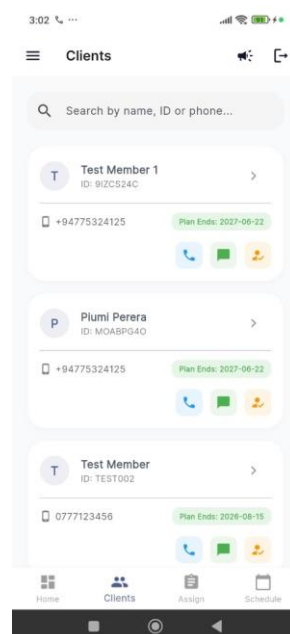
3.4.14 Personal Trainer Hub Home Dashboard

The primary operations viewport for authenticated fitness staff. It showcases daily session tallies, active client metrics, pending routine counts, a chronological global notices log, and quick navigation pathways.



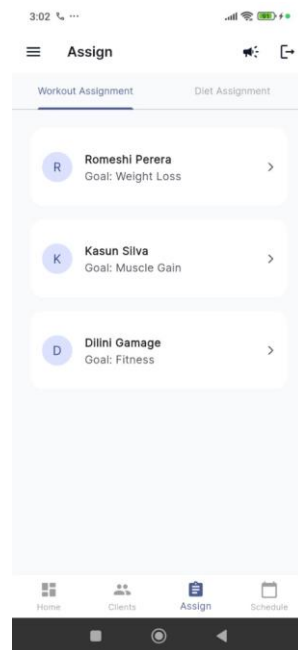
3.4.15 Personal Trainer Assigned Clients Directory

A comprehensive client management panel displaying assigned club members. The system structures dedicated status cards with user names, registration indexes, direct communication triggers, and membership validation periods.



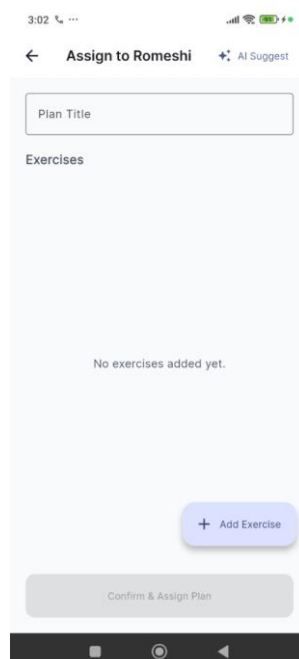
3.4.16 Goal-Oriented Workout Assignment Roster

A streamlined navigation interface that groups active athletes based on their selected health milestones (such as weight loss, muscle gain, or general physical fitness) to help instructors assign targeted training tracks.



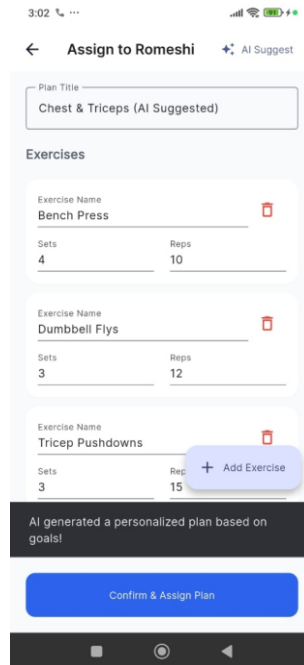
3.4.17 Custom Exercise Routine Building Canvas

An interactive form enabling trainers to construct individual fitness programs from scratch. It features clear title fields alongside an extension module to manually bind individual movements to a client's plan.



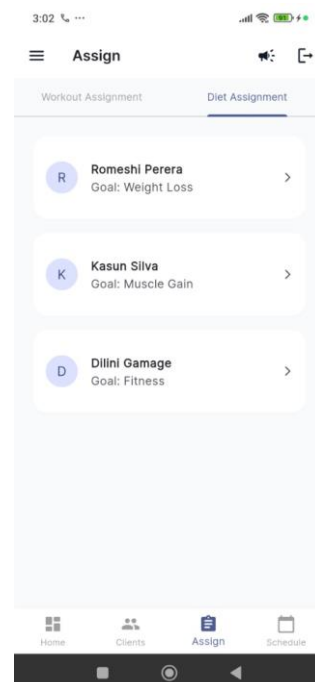
3.4.18 AI-Suggested Workout Matrix Validation Canvas

An advanced optimization module displaying structured exercise sets and repetition parameters pre-compiled by the automated backend system. It allows coaches to audit and modify the matrix before saving it.



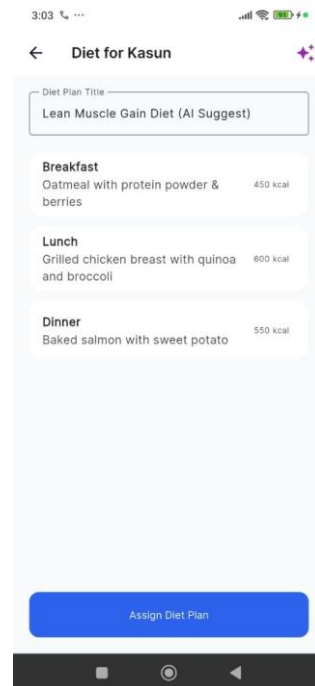
3.4.19 Goal-Oriented Dietary Assignment Roster

An instructor-side overview page that filters and tracks assigned gym members by their metabolic and physiological targets, providing a gateway to deploy tailored nutritional strategies.



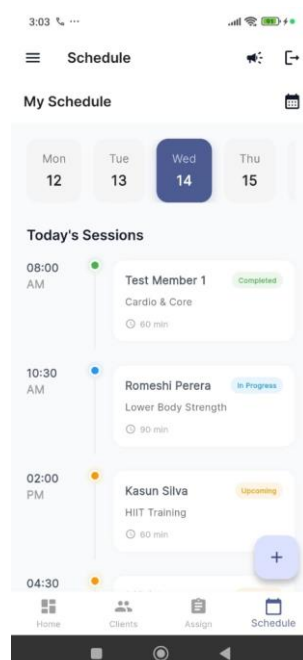
3.4.20 AI-Suggested Dietary Target Matrix Preview

A nutritional visualization viewport detailing automated dietary advice generated by the system. It breaks down food selection sets and precise caloric milestones across localized daytime phases.



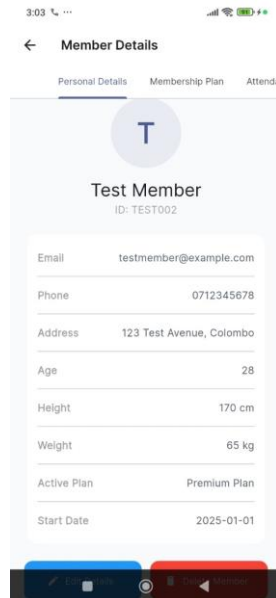
3.4.21 Personal Trainer Master Booking Schedule

An interactive calendar module structuring an instructor's training timeline. It categorizes client appointments into colored tags for completed, in-progress, or upcoming slots to prevent scheduling clashes.



3.4.22 Detailed Club Member Biographical Profile

A granular human resource data view that aggregates a member's personal communication links, residential address records, age metrics, current height and weight markers, and initialization dates.

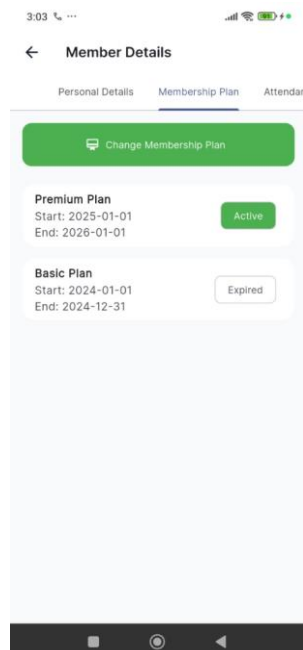


The screenshot shows the 'Member Details' screen with the 'Personal Details' tab selected. It features a profile card for 'Test Member' (ID: TEST002) with a circular avatar containing the letter 'T'. Below the card is a form with the following fields:

Email	testmember@example.com
Phone	0712345678
Address	123 Test Avenue, Colombo
Age	28
Height	170 cm
Weight	65 kg
Active Plan	Premium Plan
Start Date	2025-01-01

3.4.23 Club Member Subscription Plan Management View

A dedicated subscription panel showing a user's active or expired membership history cards, with an integrated administrative routing shortcut to update or change package allocations.

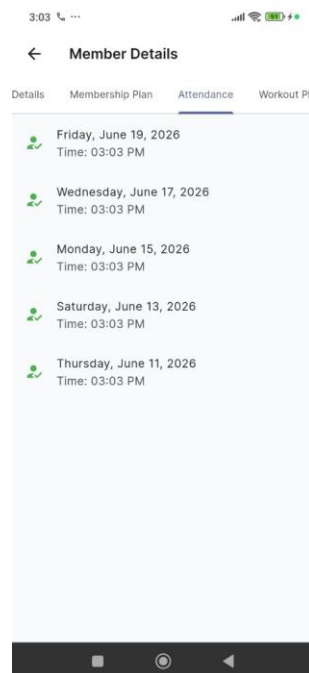


The screenshot shows the 'Member Details' screen with the 'Membership Plan' tab selected. It features a green button labeled 'Change Membership Plan' at the top. Below it are two membership plan cards:

Plan Name	Start Date	End Date	Status
Premium Plan	2025-01-01	2026-01-01	Active
Basic Plan	2024-01-01	2024-12-31	Expired

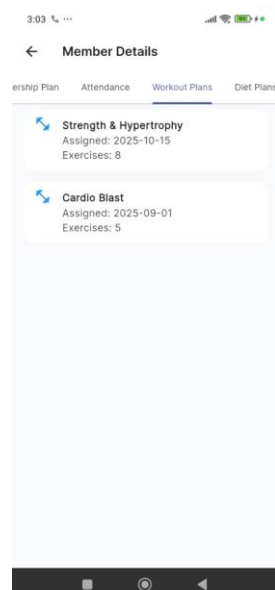
3.4.24 Club Member Attendance History Tab

An indexing sub-view that structures a chronological log of a member's facility check-ins. The tab populates clear date stamps and specific arrival time indicators, confirming long-term physical consistency parameters for system evaluators and trainers.



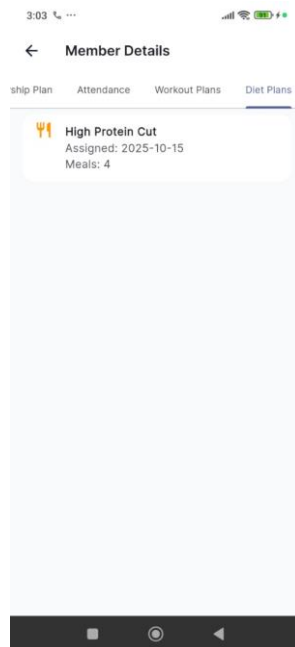
3.4.25 Member Assigned Workout Plans Feed

A history viewport displaying active or past physical routines bound to an athlete's profile. Each log card tracks distinct program nomenclature (such as strength training or cardiovascular conditioning), the official assignment date, and the total metric count of individual exercises included in the track.



3.4.26 Member Dietary Target Plans Feed

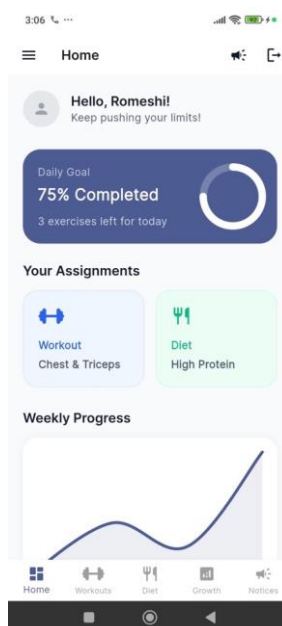
A dedicated tracking interface that populates personalized nutritional strategies bound to a member profile. The card details structural dietary plan names, assignment timeline metrics, and the aggregate quantity of daily meal targets established by the instructor.



Member Dashboard View Interfaces

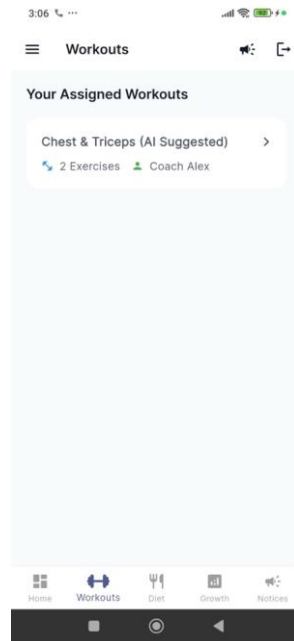
3.4.27 Gym Member Landing Hub Workspace

The primary entry landing screen for members. It displays an interactive dial tracking completion percentages for the day's tasks, dedicated shortcuts to assigned routines, and a weekly performance trend line.



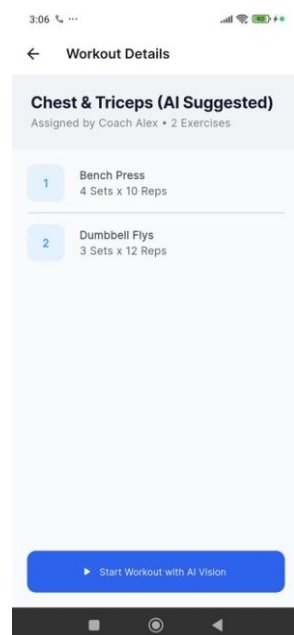
3.4.28 Member Assigned Workouts Registry

A client-side navigation screen listing active physical routines. The view maps out targeted training packages alongside summary details showing the aggregate quantity of exercises and the specific personal trainer assigned to the routine.



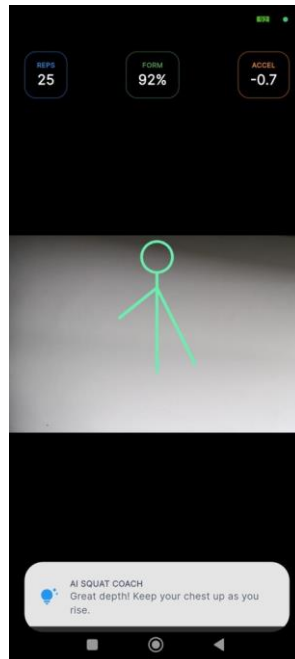
3.4.29 Workout Routine Activity Breakdown

A detailed tracking view mapping out exercise breakdowns, detailing targeted set counts and repetition splits. It features a primary action button to boot up the camera-driven computer vision validation system.



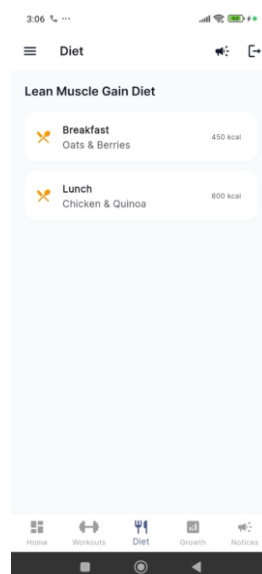
3.4.30 Interactive AI Squat Coach View

A live camera view running computer vision models locally on the mobile terminal. It maps body skeleton tracks to count repetitions, flags form accuracy configurations, monitors accelerometer tracking parameters, and outputs instant coaching tips.



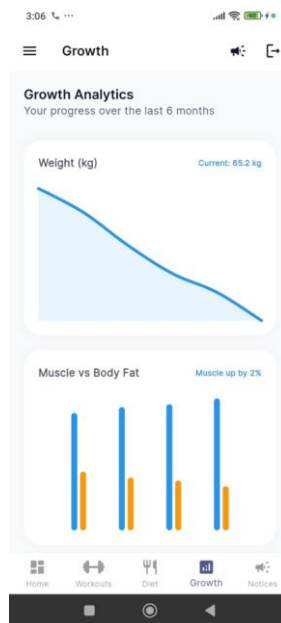
3.4.31 Dietary Program Nutritional Target Feed

A nutritional tracking viewport rendering personalized dietary strategies. The panel structures explicit food selections alongside target caloric limits across separate phases of the member's daily routine.



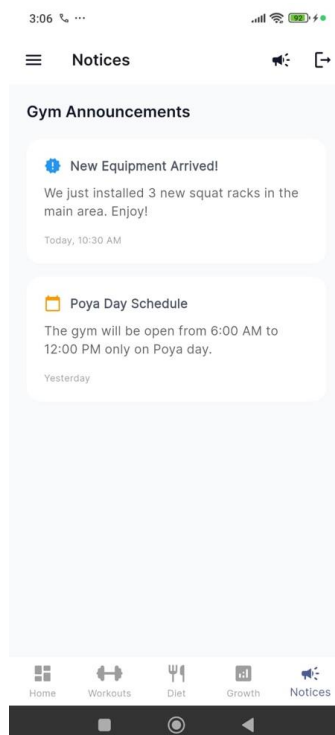
3.4.32 Member Growth Analytics Dashboard

An analytical metrics viewport tracking individual physiological indicators over time. It renders active weight fluctuation tracking lines alongside side-by-side bar charts tracking long-term changes in muscle mass and body fat percentages.



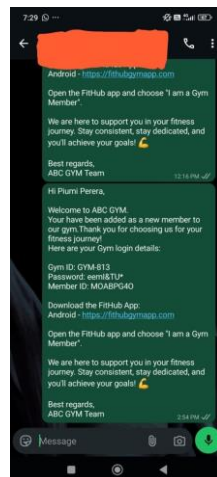
3.4.33 Gym Announcements Feed Panel

A client-side notification panel displaying real-time administrative messages. It updates instantly to keep users informed about facility upgrades, holiday hour re-alignments, and scheduling revisions.

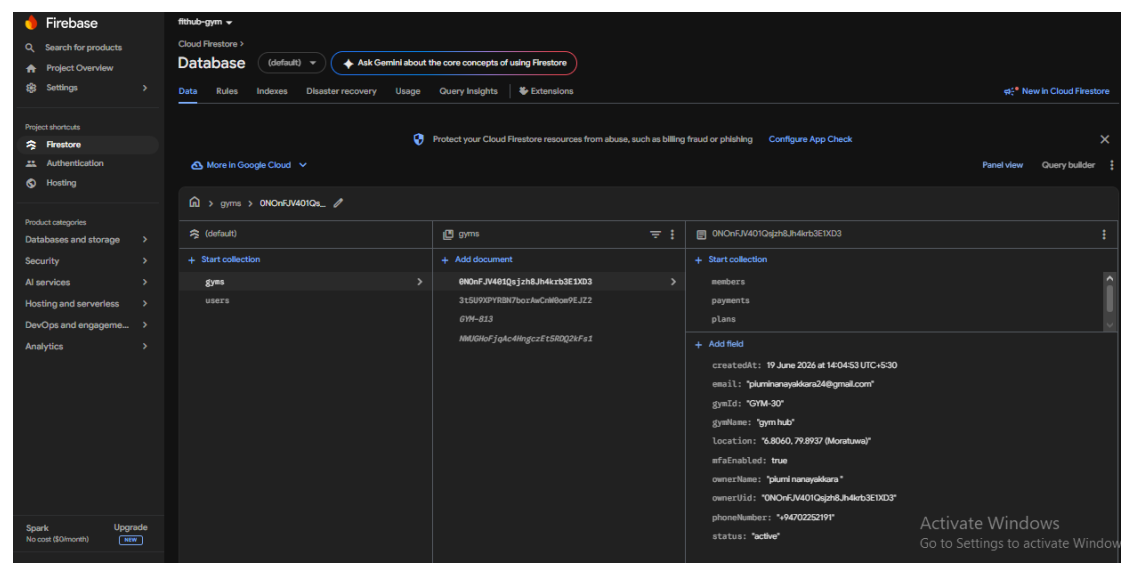


3.4.34 Automated WhatsApp Onboarding Credential Notification Subsystem

This external notification pipeline triggers automatically upon the registration of a new member or staff profile. It dispatches a secure, real-time messaging payload directly to the user's verified device, delivering the generated **Gym ID**, **Member ID**, and **Password** required for initial system authentication.



Database



Search for products

Project Overview

Settings

Project shortcuts

Authentication

Firestore

Hosting

Product categories

Databases and storage

Security

AI services

Hosting and serverless

DevOps and engage...

Analytics

Spark

No cost (\$0/month)

Upgrade

NEW

fttuh-gym

Authentication

Users | Sign-in method | Templates | Usage | Settings | Extensions

The following authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number or user UID

Add user

Identifier	Providers	Created	Signed in	User UID
mnik@gmail.com		19 Jun 2026	19 Jun 2026	MCA2PG4oErRfhgthC6Bu...
plunipramody22@gmail...		19 Jun 2026	19 Jun 2026	4m5V8dLdeQDn8Cz634He...
jph@gmail.com		19 Jun 2026	19 Jun 2026	wJWCgqvUJ2qRWZvg58L...
jlg@gmail.com		19 Jun 2026	19 Jun 2026	bgYvwyk2CnYH0NCMCMS...
vghhh@gmail.com		19 Jun 2026	19 Jun 2026	IL5VRGmG3z14dRhwgthn...
upelgraphic@gmail.com		19 Jun 2026	19 Jun 2026	bfndP8nUMZdwT8Ap9gy20...
pluminaryakana24p...+94702252191		19 Jun 2026	19 Jun 2026	ONOrFJW47DQqjh8Jh4rb3E...
2001riversmurya@g...		19 Jun 2026	19 Jun 2026	KGMyCpQJuxXXPzGO027...
hyn@gmail.com		19 Jun 2026	19 Jun 2026	KL5d7YndS3UTo0554euz...
dgh@gmail.com		19 Jun 2026	19 Jun 2026	92C24CKrMDownlyPCMAK...
xyn@gmail.com		19 Jun 2026	19 Jun 2026	NK03MNU3hQB8kshwQBFg...
abc@gmail.com		19 Jun 2026	19 Jun 2026	Muudm1QdvQWf3ARbjZofa...

Activate Windows
Go to Settings to activate Windows

CHAPTER 04: METHODOLOGY AND IMPLEMENTATION

4.1 Development Methodology

The system was developed using the Agile Scrum development process. In engineering cycles, two weeks were set aside for achieving incremental builds. Early versions concentrated on the main services such as the definition of database rules and pipelines of user tokens. Subsequent sprints covered presentation layers, automated generation scripts, and integration of the edge computer vision. Sprints ended with the compilation testing on target mobile devices to guarantee the stability and performance of the system.

4.2 Technologies and Tools Used

The deployment stack relies on production-grade, open-source software tools:

- **Client UI Layer Engine: Flutter SDK (Dart Framework)** compiles highly performant native application footprints from a single unified codebase.
- **Distributed Cloud Architecture: Firebase System Suite** manages multi-role authentication vectors and scalable JSON document storage.
- **Data Plotting Pipeline: FL Chart Core Engine** maps complex biometrics tables into responsive, lightweight line and bar graphs.
- **Version Governance Ecosystem: Git Platform & GitHub Desktop** coordinates isolated branch development, tracks file changes, and manages code history snapshots.

4.3 Future Implementation

The computer vision pipelines will be moving from the specific devices to cloud processing networks, starting with future versions and upgrades. This modification will enable a system to perform an analysis of complex athletic movements without being constrained by consumer mobile hardware barriers. Also, integrated payment systems will support the processing of international subscription plans directly in secure checkout modules.

CHAPTER 05: PLATFORM GOVERNANCE AND DELIVERY

5.1 Individual Contribution Statement

The project members below completed the structural engineering, system architecture design, and cloud deployment pipelines of the application ecosystem in the FitHub project as a collaborative group capstone project.

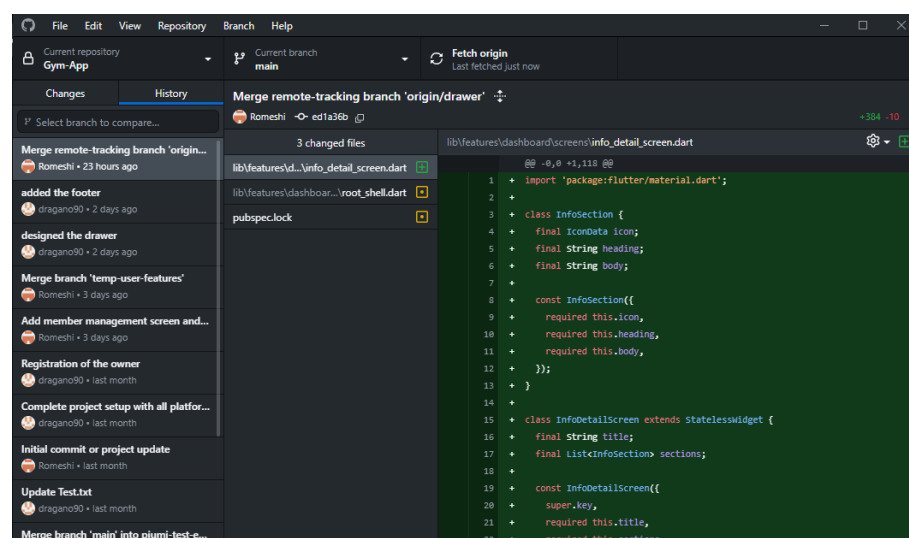
- **Piumi Nanayakkara (Student Index: 29366):** Owns the core database security schema definitions, is responsible for the relational structure of the documents within the Cloud Firestore backend, develops dynamic views for clients using the Flutter framework, maintains system version control management configuration and tracks deployments to the production environment across all active functional systems.
- **Romeshi Warnasuriya (Student Index: 28009):** In charge of implementing backend business logic modules to manage user and staff authentication as well as registration workflow, database transactional pipelines for owner-level admin tasks and getting designs of the structural layout of the user interface, along with styling aspects of the presentation of the client-side screen.

5.2 GitHub Commit History

Project development operations, codebase revisions, and incremental feature updates were systematically tracked through an isolated branch architecture using Git version control management

- **Official Remote GitHub Repository URL Link:**

<https://github.com/Romeshi/Gym-App.git>



5.3 Project Source Code Deployment Vector

The full engineering source folder, configuration files, and relational script setups have been exported to the institution's cloud server. The deployment can be reviewed via the secure access routes listed below:

- **Secure Institutional Source Code Directory Link (Uncompressed Archive):**
[https://nsbm365-my.sharepoint.com/:f:/g/personal/mrkwarnasuriya_students_nsbm_ac_lk/IgDVUs_-91NfRZGoF7epkFFVAW3q5L-wvC-BACkvVroIpk4?e=dX2xKG](https://nsbm365-my.sharepoint.com/:f:/g/personal/mrkwarnasuriya_students_nsbm_ac_lk/IgDVUs_-91NfRZGoF7epkFFVAW3q5L-wvC-BACkvVroIpk4?e=dX2xKG)
- **Secure Institutional Project Deployment Package Link (Compressed ZIP Archive):**
[https://nsbm365-my.sharepoint.com/:u:/g/personal/mrkwarnasuriya_students_nsbm_ac_lk/IQBfRC8TrYy4QZR6Eth2UkSqAR_7osm---1KiCxcOPH0zYI?e=Aj3ZYh](https://nsbm365-my.sharepoint.com/:u:/g/personal/mrkwarnasuriya_students_nsbm_ac_lk/IQBfRC8TrYy4QZR6Eth2UkSqAR_7osm---1KiCxcOPH0zYI?e=Aj3ZYh)

REFERENCE LIST

- Sommerville, I., 2015. *Software Engineering*. 10th ed. Boston: Pearson.
- Fowler, M., 2018. *Refactoring: Improving the Design of Existing Code*. 2nd ed. Boston: Addison-Wesley.
- Firebase Documentation, 2026. *Cloud Firestore Security Rules Architecture Protocols*. [online]

Available at: <https://firebase.google.com/docs/firestore/security> [Accessed 16 June 2026].

- Flutter API Reference, 2026. *Material Design App Architecture Component Specifications*. [online] Available at: <https://api.flutter.dev> [Accessed 18 June 2026].